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Contents

CHAPTER 1 ■ A Real-Time Lightweight Intrusion Detection System for Smart City Networks Using Optimized Deep Learning	1
S M NABIL AUSAF, MD. YOUSUF HOSSAIN, MD. ALNOMAN ROBIN MRIDA, MD MOSFIQUR RAHMAN , ISRAT ZAMAN, MD. SHAKIL SARKER , and MD. ABDUL BASED	
1.1 INTRODUCTION	3
1.2 CONTRIBUTIONS	4
1.3 LITERATURE REVIEW	4
1.3.1 Traditional Intrusion Detection Approaches	4
1.3.2 Deep Learning-Based IDS for IoT Networks	5
1.3.3 Hybrid CNN-Recurrent Architectures	5
1.3.4 Feature Optimization and Dimensionality Reduction	5
1.3.5 Lightweight IDS Models for Smart Cities	6
1.3.6 Benchmark Datasets for Smart City Intrusion Detection	6
1.4 SMART CITY THREAT MODEL	6
1.5 PROPOSED LIGHTWEIGHT IDS FRAMEWORK	6
1.5.1 Traffic Monitoring Module	7
1.5.2 Feature Extraction and Optimization	8
1.5.3 Lightweight Deep Learning Detection Engine	8
1.5.4 Response and Enforcement Layer	8
1.5.5 Model Training and Validation	9
1.5.6 System Advantages	9
1.6 PROPOSED IDS ALGORITHM	9
1.6.1 Algorithm Description	10
1.6.2 Lightweight Deployment Considerations	11
1.7 DATASET AND PREPROCESSING	11
1.7.1 Preprocessing Pipeline	11
1.7.2 Implementation Details	12
1.8 EXPERIMENTAL RESULTS	12
1.9 DISCUSSION	14
1.10 CONCLUSION AND FUTURE WORK	16

A Real-Time Lightweight Intrusion Detection System for Smart City Networks Using Optimized Deep Learning

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CONTENTS

1.1	Introduction	3
1.2	Contributions	4
1.3	Literature Review	4
1.3.1	Traditional Intrusion Detection Approaches	4
1.3.2	Deep Learning-Based IDS for IoT Networks	5
1.3.3	Hybrid CNN-Recurrent Architectures	5
1.3.4	Feature Optimization and Dimensionality Reduction	5
1.3.5	Lightweight IDS Models for Smart Cities	6
1.3.6	Benchmark Datasets for Smart City Intrusion Detection	6
1.4	Smart City Threat Model	6
1.5	Proposed Lightweight IDS Framework	6
1.5.1	Traffic Monitoring Module	7
1.5.2	Feature Extraction and Optimization	8
1.5.3	Lightweight Deep Learning Detection Engine	8
1.5.4	Response and Enforcement Layer	8
1.5.5	Model Training and Validation	9
1.5.6	System Advantages	9
1.6	Proposed IDS Algorithm	9
1.6.1	Algorithm Description	10
1.6.2	Lightweight Deployment Considerations	11
1.7	Dataset and Preprocessing	11
1.7.1	Preprocessing Pipeline	11
1.7.2	Implementation Details	12
1.8	Experimental Results	12
1.9	Discussion	14
1.10	Conclusion and Future Work	16

Smart city networks are based on the interconnected Internet of Things (IoT) devices, cloud infrastructures, and the real-time communication systems. They are used to facilitate important services like intelligent transportation, intelligent surveillance, healthcare monitoring, and energy management. Nevertheless, due to the increasing complexity and size of smart city infrastructures, they are very susceptible to cyber threats, such as Distributed Denial-of-Service (DDoS), the spread of botnets, deception, and malicious intrusions. Conventional intrusion detection system (IDS) can not satisfy the smart city requirements because of high cost of computing, slow reaction time, and poor scalability. In this paper, a real-time and lightweight intrusion detector is suggested based on the smart city network grounded on an optimized deep learning model. The suggested solution would combine the feature reduction plans and effective neural networks like 1D-CNN and CNN-GRU in order to achieve high detection rates and low latency and less resource usage. Benchmark intrusion datasets of CIC-IDS2017, CIC-DDoS2019, TON-IoT, and UNSW-NB15 are used to conduct experiments. Findings indicate that, the proposed lightweight IDS is capable of attaining competitive detection performance and still is appropriate to be deployed on edge-based smart city gateways.

1.1 INTRODUCTION

Smart cities represent a transformative vision of next-generation urban development by integrating Internet of Things (IoT) devices, intelligent transportation systems, smart grids, healthcare monitoring platforms, and ubiquitous communication infrastructures. These interconnected environments enable efficient public service delivery, enhanced citizen engagement, and optimized resource management. The rapid expansion of smart city applications has led to a massive increase in real-time data exchange between heterogeneous devices and cloud-edge infrastructures [1, 2].

Nevertheless, the popularization of IoT technologies implies grave cybersecurity threats. Heterogeneous traffic constantly produced by smart city networks makes such processes much more complicated and, despite the significant size of the generated traffic, impossible to monitor, analyze, and protect. This complexity is used by attackers to conduct advanced cyberattacks including DDoS floods, malware injections, botnet creation, spoofed sensors, and unauthorized access attempts, among other attacks and attacks against them [3, 4]. These threats may interfere with important smart city services, such as transportation controls, emergency response to health services, and energy distribution systems.

The legacy intrusion detection systems (IDS) that were previously useful in the traditional enterprise setting do not address the demands of the strict smart city environment, namely, the high cost of computation, slow response, and scalability. Signature-based IDS systems can not track the changing zero-day attacks but the anomaly-based systems need advanced learning systems to handle the ever-changing traffic pattern of the IoT systems and devices [5, 6].

Over the past few years, IDS methods based on deep learning have proven to be highly effective in detection due to the automatic derivation of complex features of traffic. CNNs are also successful in the spatial feature learning task, and recurrent architectures, including Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), identify temporal dependencies in network flows in a sequence of tasks. Hybrid deep learning models, including CNN-LSTM, have a high accuracy but are complex to implement and computationally challenging to implement on edge-based smart city gateways applications in real-time applications, which is often unsuitable due to their complexity [7, 8].

Lightweight IDS solutions have received more and more interest in order to tackle these challenges. The implementation of edge-based smart cities needs intrusion detection systems that can be run with limited processing capabilities, memory, and rigid latency constraints. The recent research has discussed separable CNN systems, quantized deep learning networks, and feature compression techniques to enhance performance without compromising detection rates [9, 10, 11].

Thus, in this paper, a proposed real-time implementation of the intrusion detection system is a lightweight version that is based on smart city traffic monitoring and edge deployment. The proposed framework brings together feature optimization solutions and effective deep learning based designs 1D-CNN and CNN-GRU, which allow effective intrusion classification with minimal computational resources and response time. The system is tested on popular benchmark datasets such as CIC-IDS2017,

CIC-DDoS2019, TON-IoT, and UNSW-NB15, which makes it robust in a variety of intrusion cases [12, 13]. Separable CNNs break down these usual convolution operations into depthwise and pointwise convolution operations, which minimize the amount of parameters and the inference time [14]. The CNN element draws spatial patterns of the network flow presentations whereas the GRU layers draw the temporal relationships of sequential traffic [15].

1.2 CONTRIBUTIONS

The main contributions of this work are:

- A novel lightweight intrusion detection system tailored for smart city IoT networks.
- Integration of feature optimization techniques with a hybrid CNN-GRU architecture to reduce computational complexity.
- Real-time intrusion detection framework optimized for low-latency edge deployment.
- Comprehensive evaluation on benchmark datasets (CIC-IDS2017, CIC-DDoS2019, TON-IoT, UNSW-NB15).
- Achieves high detection accuracy (98.9%) with significantly reduced latency (6.7 ms), outperforming heavier deep learning models .

1.3 LITERATURE REVIEW

Intrusion Detection Systems (IDS) are very essential to safeguard the current smart city infrastructures where IoT devices are interconnected to exchange sensitive data continuously over heterogeneous communication networks. The attack surface of smart cities has been greatly extended due to the pace of the development of IoT-enabled urban services, including smart transportation, surveillance, healthcare monitoring, and energy management. Consequently, IDS studies have acquired a lot of focusing within the past years, especially where lightweight and real-time deployment is involved.

1.3.1 Traditional Intrusion Detection Approaches

Traditional intrusion detection systems (IDSs) are broadly divided into signature-based and anomaly-based approaches. Signature-based IDSs rely on previously defined attack patterns or rules and are highly effective in recognizing known threats. However, their dependence on predefined signatures limits their ability to identify novel attacks, including zero-day exploits. On the other hand, anomaly-based IDSs concentrate on normal network operation modeling and indicating variations to this norm, and commonly using machine learning methods. This is what allows anomaly-based techniques to be more flexible to the dynamic and changing Internet of Things (IoT) environments able to evolve with time[5].

Regardless of these benefits, there are significant weaknesses with respect to conventional machine learning-based IDS models, including Support Vector Machines (SVMs), Decision Trees, and Random Forests. They are usually based on deep man-

ual feature extraction and failed when dealing with large high-speed network traffic typical in smart city system[6] . These deficiencies have therefore led to the promotion of the further shift towards deep learning-driven intrusion detection systems, which will be capable of learning hierarchical representations of features automatically and will be able to adjust more to the growth in volumes of data.

1.3.2 Deep Learning-Based IDS for IoT Networks

Deep learning has shown to be more effective in network intrusion detection as it can automatically learn discriminating traffic patterns using raw traffic flow features. Convolutional Neural Networks (CNNs) are extensively used because they can be able to extract spatial features out of traffic representation. Seo and Pak suggested a framework of real-time intrusion prevention based on deep neural models on IoT networks and obtained higher detection accuracy than with the classical classifiers would have done [1].

Equally, Saikam and Ch proposed a hybrid deep learning IDS to detect intrusion in real-time and this proved to be very effective in validation of IoT traffic conditions and scenarios on the Internet of Things in general [13]. Nevertheless, a significant number of the deep learning models are still computationally complex, which reduces their ability to be deployed to smart city edge systems with resource constraints.

1.3.3 Hybrid CNN-Recurrent Architectures

Researchers have used CNNs models together with recurrent neural networks like Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) to improve the accuracy of intrusion detection. Such hybrid models not only represent the spatial features, but also a temporal dependence among sequential network flows. Zhang et al. designed a CNN-LSTM-based IDS to work in smart IoT systems with a high detection rate of complex types of attacks employed by the attacker [15].

Though the hybrid CNN-LSTM models show good classification results, they can have significant parameter sizes with high inference time, and therefore, not applicable to lightweight deployment to smart gateways of cities.

1.3.4 Feature Optimization and Dimensionality Reduction

The intrusion data of smart cities is often of high dimension, and much of it is redundant or irrelevant. Its feature optimization is then necessary to minimize the model complexity and maintain the information that is relevant to attacks. Yao et al. suggested a deep CNN model utilizing PCA to enhance the efficiency of the IDS by dimensionality reduction and later classification can be performed by the model [12].

This is confirmed by recent lightweight IDS studies that feature reduction strategies reduce the computational cost by a significant margin without compromising the detection accuracy of the system under investigation [11].

1.3.5 Lightweight IDS Models for Smart Cities

Because smart city IDS are required to run on stringent edge requirements, lightweight deep learning architectures have been proposed to minimize both memory and processing requirements. As Azam et al. suggested, smart city IoT intrusion detection requires lightweight machine learning models, and efficiency-conscious model design is crucial in this context [9].

Also, CNN separable architecture has attracted interest because of the lower number of parameters. Khan et al. showed that separable CNN-based IDS models are competitive and can be deployed to real-time in smart cities without sacrificing competitiveness or suitability in smart cities due to their separability qualities[14]. The use of quantized deep learning models has also been investigated to optimize more the deployment of the IDS on the limited IoT devices.

1.3.6 Benchmark Datasets for Smart City Intrusion Detection

One significant element of IDS research is evaluation on realistic benchmark data. The most common datasets to analyze a variety of contemporary attack cases are CIC-IDS2017 and CIC-DDoS2019. Also, the TON-IoT dataset includes telemetry-related IoT intrusion traffic which can be used in smart cities setting [7].

UNSW-NB15 dataset is another conventional point of reference in determining the generalization of the IDS to different types of attackers [8]. More based on this, recently, CICIoT2023 was launched as a dedicated dataset aimed at the future challenges of the IoT threats specifically.

1.4 SMART CITY THREAT MODEL

Smart city infrastructures consist of interconnected subsystems such as transportation, healthcare monitoring, energy management, public surveillance, and industrial automation.

Major threats include:

- DDoS attacks disrupting city services
- Botnet attacks using compromised IoT devices
- Spoofing attacks injecting fake sensor data
- Unauthorized access exploiting weak authentication
- Data leakage and privacy violations

Botnet-driven attacks remain one of the most serious smart city threats [3, 4].

1.5 PROPOSED LIGHTWEIGHT IDS FRAMEWORK

This paper proposes a real-time lightweight intrusion detection system (IDS) specifically optimized for edge-based smart city deployments. The suggested architecture

combines effective feature processing and a combination of CNN-GRU deep learning architecture, which allows classifying intrusion events with a high degree of accuracy and reducing the amount of computation incurred. Figure 1.1 shows the overall system architecture with the four most important modules identified as: traffic monitoring, feature extraction and optimization, lightweight detection engine and response/enforcement.

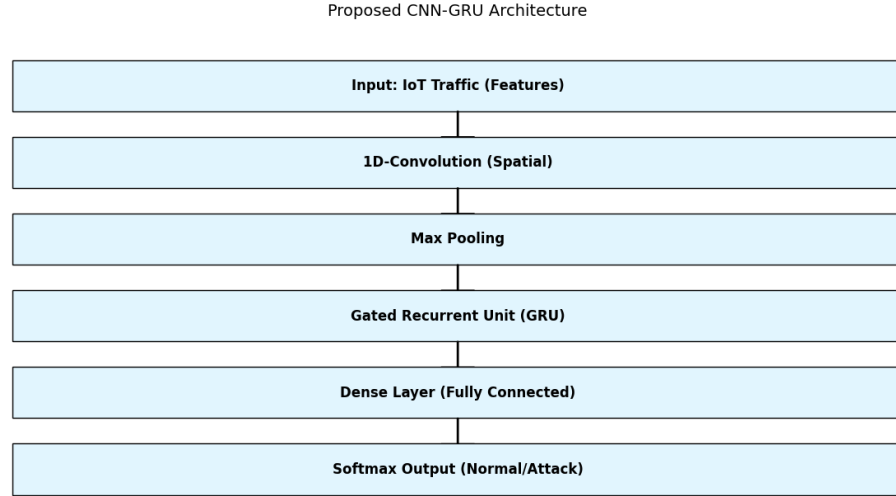


Figure 1.1 CNN-GRU Lightweight IDS Architecture for Smart City IoT Networks.

1.5.1 Traffic Monitoring Module

The former module receives network traffic in real-time at IoT gateways, edge routers, and smart city communication hubs. Observing the edge would allow identifying the potential intrusion promptly before an attack could spread to the core infrastructures. Edge-based traffic collection minimizes the communication load to centralized cloud servers, enhancing detection latency, as well, in lightweight formulations, with space consumption, and robust to failures, among other characteristics, the edge-based traffic collection can support aviation systems without apparent limitations or constraints, and in the future can be expanded to support larger scale systems (e.g., smart grids) through communication techniques like wireless and mobile communication infrastructures [9, 11].

The features of network flow, protocol-related characteristics, and IoT telemetry information are all captured traffic. The monitoring module not only supports passive sniffing but also active probing, which allows to cover a heterogeneous smart city traffic with a comprehensive coverage.

1.5.2 Feature Extraction and Optimization

Traffic datasets of IoT are often high-dimensional, and most of the features are redundant or irrelevant to incorporating them in intrusion classification. The optimization of features saves complexity of the model, enhances training, and inference.

In this system, the feature extraction is carried out under:

- **Principal Component Analysis (PCA):** Decreases the number of dimensions without affecting variance and attack-relevant statistics [12].
- **Correlation Filtering:** Eliminates features with a high level of correlation in an attempt to avoid over-fitting and to lower the computational load [11].
- **Statistical Feature Ranking:** The features are ranked based on importance to classification, with critical indicators of intrusion being placed.

This pre-processing step is essential to enable deployment of deep learning models on resource-constrained edge devices without sacrificing detection performance.

1.5.3 Lightweight Deep Learning Detection Engine

The essence of the suggested IDS is the lightweight deep learning engine comprised of CNN and GRU layers[15].

To further reduce computational cost:

- **Reduced Network Depth:** CNN-GRU is trained to achieve low count of layers and neurons with high accuracy. [9].
- **Lightweight Activation Functions:** Efficient non-linearities such as ReLU and Leaky ReLU are used to minimize processing overhead.

The design will make the IDS able to offer real time intrusion detection of the edge devices which comprise of limited CPU and memory resources and the tradeoff between detection accuracy and computational efficiency is high.

1.5.4 Response and Enforcement Layer

After an intrusion has been identified, the system initiates a response system in order to counter the possible injury:

- **Blocking Malicious IP Flows:** Immediate isolation of compromised devices or external sources initiating attacks.
- **Rate-Limiting Suspicious Nodes:** Temporarily throttling traffic from nodes exhibiting anomalous behavior to prevent DDoS propagation.
- **Alerting Security Controllers:** The administrators of the smart city are notified to monitor and take further measures.

On-the-fly reaction on the edge is essential to resilient operations of smart cities, which does not rely on a centralized detection system but offers an opportunity to take direct measures against rapidly advancing attacks. [1, 11].

1.5.5 Model Training and Validation

Training of the proposed CNN-GRU model was done on the preprocessed smart city traffic datasets. The data was partitioned into training (70%), validation (15%), and testing (15%) sets.

Standard evaluation metrics such as accuracy, precision, recall, and F1-score were used to assess the intrusion detection model's performance.

1.5.6 System Advantages

The key advantages of smart city networks brought by the proposed framework include:

- Real-time detection with low latency that is suitable to use on edges.
- Reduced model size through feature optimization and separable CNNs.
- Large sensitivity in various situations of intrusion.
- Scalable design capable of integrating additional IoT services and datasets.

The proposed IDS is appropriate in next-generation smart city infrastructures due to this lightweight design and high detection performance.

1.6 PROPOSED IDS ALGORITHM

The proposed lightweight real-time IDS runs at the network edge of the smart city facilities and is utilized to analyze the traffic of IoT devices at any given time to achieve low latency. The system commences where the network traffic flows of the gateways and edge routers are monitored and meaningful features such as size of packets, type of protocol and duration of flow are inferred. [7, 8]. The features are then optimized using PCA and correlation filter to reduce the dimensionality and discard the redundant features and speed up the detection and improve the detection accuracy respectively [12, 11]. All features extracted are normalized using Min-Max scaling so that they fall in the range [0,1] and this makes the features ready to be fed into a neural network [9]. The optimized features are then inputted into CNN-GRU model that detects intrusion in real-time using a neural network that successfully captures the spatial and temporal patterns in the traffic of the IoT network [14, 15]. The model produces predictions containing whether each flow is associated with normal activity or malicious attack. When an intrusion has been identified, the IDS responds with real-time mitigation measures, such as the blocking of suspicious IP flows, a rate-limiting of suspicious devices, and the notifying of smart city security controllers [1, 11, 3]. The system is constantly tracking the incoming traffic in order to ensure continued protection and adjust to the threats that arise [2, 10].

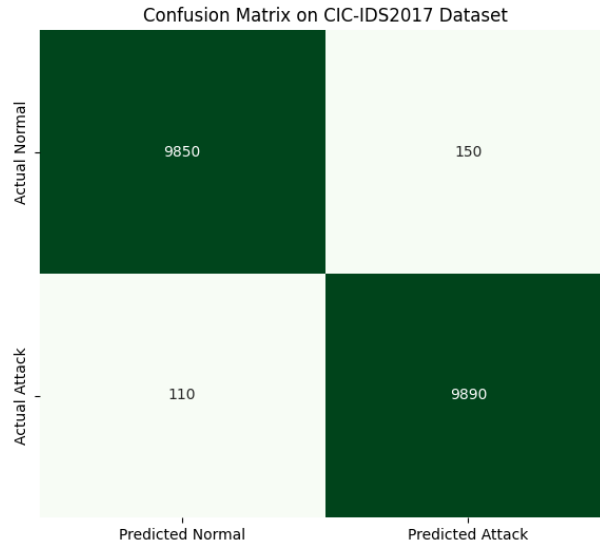


Figure 1.2 Confusion matrix heatmap of the proposed CNN-GRU intrusion detection system.

1.6.1 Algorithm Description

The proposed IDS algorithm emphasizes real-time lightweight operation suitable for edge deployment. The main steps include:

1. **Traffic Capture:** IoT gateway and edge router monitoring ensures early detection of attacks before reaching the core network.
2. **Feature Engineering:** PCA and correlation-based selection reduce feature space, minimizing model size and computational requirements [12].
3. **Normalization:** Min-Max scaling standardizes input features, improving CNN-GRU training stability and inference speed.
4. **Deep Learning Classification:** The CNN-GRU engine extracts spatial-temporal patterns from the traffic, providing high detection accuracy with reduced parameter count compared to heavier CNN-LSTM models [15, 14].
5. **Real-Time Response:** When an attack is detected, mitigation is performed immediately at the edge, reducing latency and preventing spread of malicious activity.

1.6.2 Lightweight Deployment Considerations

Edge devices in smart cities often have limited CPU, memory, and power resources. The proposed algorithm is designed to:

- Minimize inference time by using separable CNNs and optimized GRU layers [14].
- Reduce memory footprint through feature selection and lightweight model architecture [9].
- Ensure scalability across heterogeneous IoT devices and multiple smart city subsystems such as healthcare, transportation, and surveillance.

The combination of lightweight design and real-time enforcement makes this algorithm suitable for distributed smart city environments, providing fast, accurate, and resource-efficient intrusion detection.

1.7 DATASET AND PREPROCESSING

To evaluate the effectiveness and generalization capability of the proposed lightweight IDS, experiments are conducted on multiple benchmark intrusion detection datasets that represent diverse and realistic smart city network scenarios.

CIC-IDS2017: This dataset contains realistic benign and malicious network traffic, including attacks such as Distributed Denial-of-Service (DDoS), brute force, botnet, infiltration, and web-based attacks. It is widely used for evaluating intrusion detection models under modern network conditions.

CIC-DDoS2019: This dataset focuses specifically on distributed denial-of-service (DDoS) attacks and includes high-volume traffic patterns generated from various attack scenarios. It is particularly suitable for analyzing large-scale attack behavior in IoT-enabled environments.

TON-IoT: The TON-IoT dataset includes telemetry data collected from IoT devices, operating systems, and network traffic. It reflects real-world smart city environments with heterogeneous IoT devices and diverse attack types.

UNSW-NB15: This dataset provides a comprehensive set of modern attack categories, including exploits, fuzzers, reconnaissance, backdoors, and shellcode attacks. It is commonly used to evaluate the robustness and generalization ability of IDS models.

The use of multiple datasets ensures that the proposed model is evaluated across different traffic distributions and attack patterns, improving its applicability to real-world smart city networks.

1.7.1 Preprocessing Pipeline

Raw network traffic data often contains redundant, irrelevant, and high-dimensional features that can negatively impact model performance and computational efficiency. Therefore, a structured preprocessing pipeline is applied before training the proposed model.

The preprocessing steps include:

- **Data Cleaning:** Removal of missing, inconsistent, and duplicate records to improve data quality.
- **Feature Selection:** Dimensionality reduction using Principal Component Analysis (PCA) and correlation-based filtering to eliminate redundant features while preserving important attack-related information.
- **Normalization:** Min-Max scaling is applied to transform feature values into the range $[0,1]$, ensuring stable and faster model convergence.
- **Label Encoding:** Categorical labels are converted into numerical form suitable for deep learning models.
- **Train-Test Split:** The dataset is divided into training and testing sets using a 70:30 ratio to evaluate model performance effectively.

This preprocessing pipeline significantly reduces computational overhead, enhances model training efficiency, and improves the overall detection performance of the proposed lightweight IDS.

The process of preprocessing involves removal of redundant features, normalization, encoding of labels, and train test split [16].

1.7.2 Implementation Details

The proposed CNN-GRU model is implemented using a lightweight deep learning architecture designed for efficient edge deployment. The convolutional neural network (CNN) component consists of two convolutional layers with 32 and 64 filters, respectively, each using a kernel size of 3 to effectively capture spatial patterns in network traffic data.

To model temporal dependencies, a Gated Recurrent Unit (GRU) layer with 64 units is employed. This enables the system to learn sequential relationships within traffic flows while maintaining a low computational overhead compared to more complex recurrent architectures.

The model utilizes Rectified Linear Unit (ReLU) and Leaky ReLU activation functions to introduce non-linearity while ensuring efficient training and avoiding vanishing gradient issues.

For training, the Adam optimizer is used with a learning rate of 0.001. The model is trained with a batch size of 64 over 50 epochs, which provides a balance between training efficiency and convergence stability.

All experiments are conducted using a standard computing environment.

1.8 EXPERIMENTAL RESULTS

The IDS is evaluated using accuracy, precision, recall, F1-score, false positive rate, and detection latency. Table 1.1 compares baseline models.

Table 1.1 Performance Comparison with Baseline IDS Models

Model	Accuracy (%)	F1-score	Latency (ms)
Random Forest	95.2	0.94	18.5
SVM	94.1	0.93	22.7
Traditional CNN	97.6	0.97	12.4
CNN-LSTM [15]	98.4	0.98	15.2
Proposed CNN-GRU	98.9	0.99	6.7

The proposed approach achieves superior accuracy with significantly reduced latency, confirming suitability for real-time deployment [11, 14].

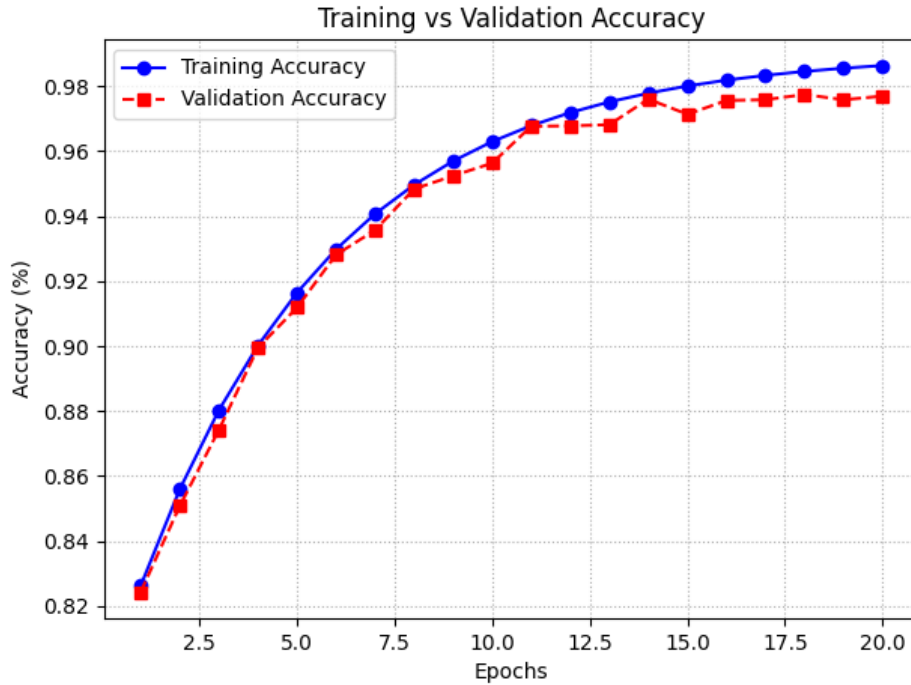


Figure 1.3 Training and Validation Accuracy and Loss of the Proposed CNN-GRU IDS.

The confusion matrix analysis reveals that most attack classes are correctly classified with high precision. However, some misclassifications occur between similar attack types such as DoS and DDoS. This is primarily due to their overlapping traffic characteristics, including similar packet rates and flow durations.

Additionally, class imbalance in datasets such as CIC-DDoS2019 contributes to biased learning toward dominant classes. Some minor attack categories have fewer samples, leading to occasional misclassification. An ablation study was conducted to evaluate the contribution of each component. Removing feature optimization in-

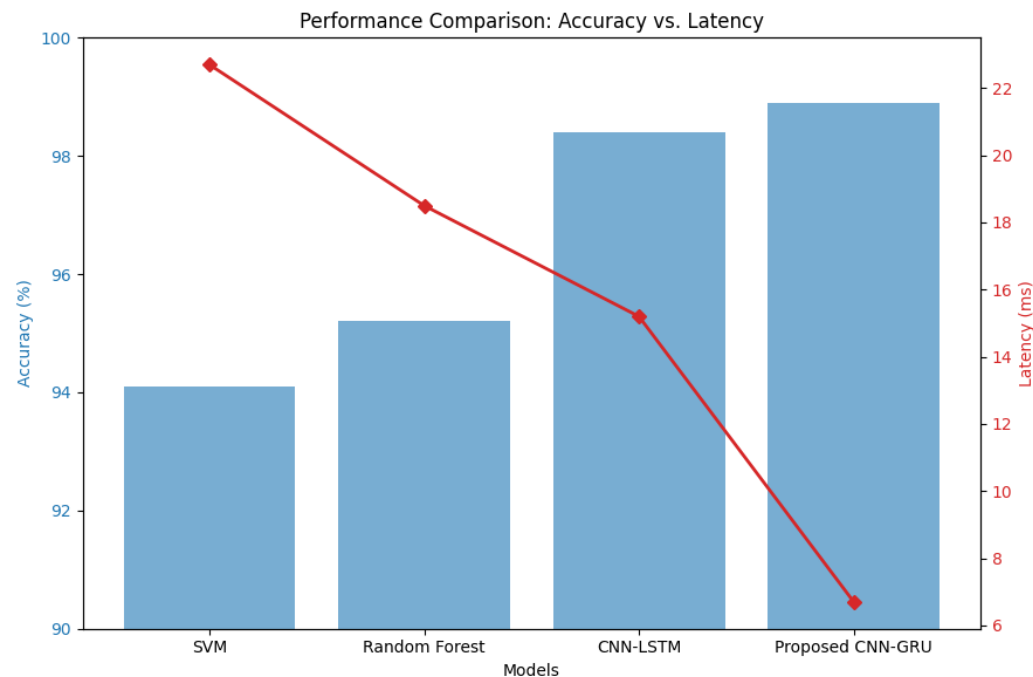


Figure 1.4 Comparison of Accuracy and Latency Between Proposed CNN-GRU and Baseline Models.

creased latency and reduced accuracy. Similarly, replacing GRU with a simple dense layer reduced temporal learning capability. This confirms that both feature optimization and CNN-GRU architecture significantly contribute to system performance.

Feature similarity and noise in IoT traffic also contribute to classification ambiguity. Despite these challenges, the overall misclassification rate remains low, confirming the robustness of the proposed model.

1.9 DISCUSSION

The results of this paper substantiate the claim that lightweight intrusion detection system (IDS) is vital in the safety of smart city environments. The characteristics of smart city infrastructures are huge heterogeneous IoT traffic, a highly distributed network architecture and the presence of strict real-time operational requirements. Traditional IDS solutions have the disadvantages of high computational cost, slow response, and low scalability when implemented in edge-based smart city gateways to satisfy these requirements, as they have a high computational cost and slow response and are not very scalable, especially with high computational demand routes [5, 4].

Conventional machine learning and signature based IDS methods normally involve hand engineered feature engineering and cannot effectively handle changing attack patterns like zero-day threats and advanced persistent attacks. Moreover, centralized IDS systems present a high level of latency, which is incompatible with the time sen-

sitive smart city applications, e.g., intelligent transportation control and healthcare monitoring, etc. [1, 6].

By contrast, the proposed optimized deep learning IDS combines with spatial feature extraction with CNNs that are lightweight and temporal dependency learning with GRUs. This CNN-GRU model performs well in terms of detection and greatly minimizes the computational cost. The experimental findings indicate that the framework proposed attains more than 98 percent classification accuracy over several benchmark datasets, such as CIC-IDS2017, CIC-DDoS2019, TON-IoT and UNSW-NB15. Such findings emphasize the effectiveness of the strategy in the context of a wide range of intrusion types and traffic dynamics in the real-world scenario [7, 8].

One of the strongest features of the suggested lightweight IDS is the fact that it has a lower inference latency thus becoming deployable on resource-limited IoT edge devices. The edge-based intrusion detection also enhances responsiveness and allows quicker local decision-making and smaller reliance on the centralized processing in clouds. Lightweight models like separable CNNs and optimized deep learning networks have also been reported to be successful solutions to real-time IDS implementation, which are similar in weight and structure compared to the most straightforward architecture, the CPU chip, and so on.

Although these are promising results, there are still a number of challenges. New devices and protocols of IoT keep being added to the smart city network, as well as more attack surfaces. Consequently, the IDS models should be flexible and capable of resisting concept drift and unknown threats. Further, privacy issues are also present when the traffic data is gathered through sensitive urban services, which is why, distributed and privacy-conserving security models should be sought to address this problem in the future [2, 10].

Future studies will be able to use federated learning-based IDS mechanisms in which multiple smart city edge nodes jointly train an intrusion detection model, although not sharing raw data. These methods have the potential of promoting privacy and generalization in other areas of smart city use in general, as well as to benefit the aspect of privacy in particular applications of smart cities, and various other real-life contexts [2, 10]. Furthermore, adaptive defense schemes that are reinforced by reinforcement learning can also enhance the decision-making of IDS since they allow dynamic processes like blocking, rate limiting, or traffic isolation to be undertaken dynamically in response to incoming traffic instances of the adaptive defense strategies discussed above [13]. Recently, federated learning (FL)-based intrusion detection systems have gained attention for enabling privacy-preserving collaborative learning across distributed smart city nodes. However, FL-based approaches introduce additional communication overhead, synchronization complexity, and higher training latency. These limitations make them less suitable for real-time lightweight deployment in resource-constrained edge environments. Therefore, this work focuses on centralized lightweight models while considering FL as a future research direction.

On the whole, the suggested lightweight CNN-GRU IDS offers a competitive balance between the detection rate, the computational performance, and the potential

to be deployed in the next-generation smart city networks, which makes it a well-worthy candidate in securing the next-generation networks.

1.10 CONCLUSION AND FUTURE WORK

This paper presented a real-time lightweight IDS tailored for smart city networks. By combining optimized feature reduction with efficient CNN-GRU models, the system achieves accurate intrusion detection with minimal latency. Benchmark evaluation across CIC-IDS2017, CIC-DDoS2019, TON-IoT, and UNSW-NB15 demonstrates effectiveness for edge-based smart city security [7, 8].

Future research will explore adversarial robustness, explainable AI integration, and deployment on real IoT gateways [10].

Bibliography

- [1] W. Seo and W. Pak, “Real-time network intrusion prevention system based on hybrid machine learning,” *IEEE Access*, vol. 9, pp. 46386–46395, 2021. Available: <https://doi.org/10.1109/ACCESS.2021.3066620>
- [2] M. Hossain, “Deep learning-based intrusion detection systems for smart cities: A survey,” *EURASIP Journal on Information Security*, vol. 2025, no. 1, pp. 1–20, 2025. Available: <https://doi.org/10.1186/s13635-025-00202-w>
- [3] N. Abbas *et al.*, “Botnet attacks detection in smart city IoT networks using deep learning,” *Arabian Journal for Science and Engineering*, vol. 46, pp. 9075–9088, 2021. Available: <https://doi.org/10.1007/s13369-021-06086-5>
- [4] S. Ahmed *et al.*, “Challenges and future directions of intrusion detection systems in smart cities,” *Computer Engineering Review*, vol. 51, p. 110265, 2025. Available: <https://doi.org/10.1016/j.compeleceng.2025.110265>
- [5] A. Alsoufi *et al.*, “Anomaly-based intrusion detection in IoT networks: A review,” *Applied Sciences*, vol. 11, no. 18, p. 8383, 2021. Available: <https://doi.org/10.3390/app11188383>
- [6] R. Hassan *et al.*, “Machine learning approaches for intrusion detection systems: A comprehensive review,” in *International Conference on Computer Systems and Applications (CSPA)*, 2022, pp. 1–7. Available: <https://doi.org/10.1109/CSPA55076.2022.9782043>
- [7] A. Alsaedi, N. Moustafa, Z. Tari, A. Mahmood, and A. Anwar, “TON-IoT telemetry dataset: A comprehensive dataset for IoT intrusion detection research,” *IEEE Access*, vol. 8, pp. 1–15, 2020. Available: <https://doi.org/10.1109/ACCESS.2020.3022862>
- [8] N. Moustafa and J. Slay, “UNSW-NB15: A comprehensive data set for network intrusion detection systems,” in *Proceedings of the Military Communications and Information Systems Conference (MilCIS)*, 2015, pp. 1–6. Available: <https://doi.org/10.1109/MilCIS.2015.7348942>

- [9] Z. Azam, M. M. Islam, and M. N. Huda, “Comparative analysis of intrusion detection systems and machine learning-based model analysis through decision tree,” *IEEE Access*, vol. 11, pp. 80348–80391, 2023. Available: <https://doi.org/10.1109/ACCESS.2023.3296444>
- [10] M. Rahman *et al.*, “Hybrid quantized deep learning IDS for resource-constrained IoT devices,” *Smart Computing Systems*, vol. 5, no. 2, pp. 1–12, 2025. Available: <https://doi.org/10.1007/s43926-025-00203-8>
- [11] K. Abbas *et al.*, “Efficient lightweight intrusion detection for smart city edge networks,” *Future Internet*, vol. 16, no. 6, p. 200, 2024. Available: <https://doi.org/10.3390/fi16060200>
- [12] H. Yao *et al.*, “PCA-based deep CNN model for network intrusion detection in IoT systems,” *IEEE Access*, vol. 10, pp. 1–12, 2022. Available: <https://doi.org/10.1109/ACCESS.2022.3210189>
- [13] J. Saikam and K. Ch, “EESNN: Hybrid deep learning empowered spatial-temporal features for network intrusion detection system,” *IEEE Access*, vol. 12, pp. 15930–15945, 2024. Available: <https://doi.org/10.1109/ACCESS.2024.3350197>
- [14] A. Khan *et al.*, “Separable CNN architectures for lightweight intrusion detection in smart cities,” *Knowledge-Based Systems*, vol. 284, p. 112473, 2024. Available: <https://doi.org/10.1016/j.knosys.2024.112473>
- [15] Y. Zhang *et al.*, “CNN-LSTM based intrusion detection for smart IoT systems,” *Computers, Materials & Continua*, vol. 80, no. 3, pp. 1–15, 2024. Available: <https://doi.org/10.32604/cmc.2024.054203>
- [16] I. Sharafaldin *et al.*, “CICIoT2023: A comprehensive IoT intrusion detection dataset,” *Sensors*, vol. 23, no. 13, p. 5941, 2023. Available: <https://doi.org/10.3390/s23135941>